CLASSIFICATION: TLP:AMBER+STRICT	DEFCON 2 (ELEVATED ALERT)	TIMEZONE: IST (UTC+05:30)

CYBERTHREAT INTELLIGENCE CORRELATION DASHBOARD (CTICD)

Comprehensive Technical Project & Engineering Architecture Report

Project Version:	v2.5 Enterprise Production	**Target Environment:**	Local / Cloud Serverless (Vercel)
Author / Lead Analyst:	Vidyadhar	**Database Architecture:**	SQLite / SQLAlchemy ORM (Writable /tmp)
Host Platform:	Python 3.13 / Flask 3.1	**Standards Compliance:**	STIX 2.1, TAXII, YARA, Snort, MITRE ATT&CK;
Repository:	github.com/vidyadhar2007/CTICD	**Report Generated:**	November 2026 | Indian Standard Time

EXECUTIVE SUMMARY

The **CyberThreat Intelligence Correlation Dashboard (CTICD)** is an enterprise-grade, next-generation Security Operations Center (SOC) intelligence platform engineered to unify heterogeneous Open-Source Intelligence (OSINT), real-time sensor streams, and multi-entity graph correlation into a single operational interface. Modern enterprise defenders routinely face data fatigue caused by fragmented threat feeds, manual indicator extraction, and slow threat actor attribution. CTICD eliminates these bottlenecks by synthesizing **real-time ballistic origin-to-target attack trajectory telemetry**, automated **Diamond Model of Intrusion** graph correlation, **MITRE ATT&CK; Enterprise Matrix heatmaps**, **PE binary entropy dissection**, and continuous **Snort / YARA / firewall drop rule generation**.

This project integrates client-side hardware-accelerated rendering (MapLibre GL, Vis.js Network, Web Audio radar synthesis) with a robust Python Flask backend, RESTful telemetry pipelines, and continuous deployment configured for both on-premise dedicated SOC servers and global Vercel serverless cloud runtimes. Every telemetry event and operational clock is synchronized to **Indian Standard Time (IST - UTC+05:30)**.

1. SYSTEM ARCHITECTURE & COMPONENT TOPOLOGY

CTICD is structured into a modular, decoupled 4-tier architecture designed for low-latency ingestion, resilient correlation, and instantaneous threat intelligence presentation.

Tier / Layer	Core Technologies	Key Responsibilities & Capabilities
Presentation & Tactical HUD	HTML5, Bootstrap 5.3, MapLibre GL 4.3, Vis.js, Chart.js, Web Audio API, CSS3 Glassmorphism	Ballistic trajectory maps, Vis.js interactive correlation graph, real-time streaming attack terminal, Spotlight Command Palette (Ctrl+K), AI Copilot drawer, procedural audio radar synthesizer.
Application & Intelligence	Python 3.13, Flask 3.1, Werkzeug, ThreatEngine, GeoIPService, IOCEngine, CVEEngine, MalwareEngine	Multi-feed scoring algorithm (VT, AbuseIPDB, Shodan), Shannon entropy calculator, CVSS v3.1 vector parser, EPSS likelihood ranker, automated defense rule compiler (YARA, Snort, iptables).
Data & Persistence	SQLite 3, SQLAlchemy 2.0 ORM, Flask-Login, CSV Ingestion Engine	Relational persistence of analyst identities, security incidents, threat indicators, audit logs, and investigation case dossiers. Auto-recovering writable /tmp database for serverless environments.
Deployment & Edge Cloud	Vercel Serverless Functions (@vercel/python), Gunicorn 26, GitHub CI/CD, Waitress	Auto-build CI/CD via GitHub webhooks, Edge CDN caching for static bundles, isolated sub-second execution, zero-maintenance global availability.

2. CORE INTELLIGENCE ENGINES & TECHNICAL IMPLEMENTATION

2.1 Real-Time Geospatial Ballistic Threat Map

Built on top of MapLibre GL vector tiles, the real-time ballistic map models origin-to-destination cyber attack trajectories connecting 32 sovereign strategic hubs across 6 continents. Trajectories are computed using great-circle quadratic Bezier curves with altitude jitter. Each attack event carries millisecond-accurate timestamps in **Indian Standard Time (IST)**, source IP, ASN, targeted critical asset (e.g. 5G Standalone UPF, SCADA breakers, SWIFT gateways), and forensic raw hex packet bytes. Analysts can filter vectors dynamically (Critical C2, RCE Exploits, DDoS Floods, Auth Bypasses).

2.2 Multi-Entity Correlation Engine (Vis.js & Diamond Model)

The correlation engine maps complex, multi-stage intrusions across the **Diamond Model** (Adversary, Capability, Infrastructure, and Victim) and aligns nodes to the **Cyber Kill Chain** (Reconnaissance to Actions on Objectives). Using physics-based force-directed layouts via Vis.js Network, analysts can click and drag entities, inspect interconnected IPs, domains, hashes, and CVEs, and export automatically compiled detection signatures.

2.3 Multi-Feed IOC Search & OSINT Aggregator

Accepts IPv4, IPv6, FQDNs, URLs, MD5, SHA1, and SHA256 indicators. Cross-references multi-vendor detection matrices (VirusTotal engine ratios, AbuseIPDB confidence percentiles, AlienVault OTX, Shodan open ports). Outputs exact perimeter mitigation commands across **Linux iptables**, **Cisco IOS**, **Palo Alto PAN-OS**, and **Windows Defender Firewall**, with 1-click export to **OASIS STIX 2.1 JSON** bundles.

2.4 CVE Vulnerability & Exploit Explorer

Correlates CVE vulnerabilities (e.g. CVE-2024-3094 XZ Backdoor, CVE-2024-3400 PAN-OS RCE, CVE-2024-1709 ScreenConnect Bypass, CVE-2021-44228 Log4Shell) with CVSS v3.1 vector calculators, Exploit Prediction Scoring System (EPSS) percentiles, and CISA Known Exploited Vulnerability (KEV) catalog mandates. Automatically outputs Snort / Suricata network perimeter detection rules.

2.5 Malware Sandbox & PE Binary Entropy Dissector

Calculates static Shannon entropy per Portable Executable (PE) section (.text, .rdata, .data, .rsrc). Sections with entropy exceeding 7.0 are flagged as packed or encrypted (UPX/Themida). Dissects dynamic behavioral telemetry including hooked Win32 APIs (VirtualAllocEx, WriteProcessMemory, CreateRemoteThread), registry run keys, mutexes, and 12-vendor antivirus detections.

3. IMMERSIVE SOC WAR ROOM & ANALYST PRODUCTIVITY SUITE

Feature Module	Key Combination	Operational Functionality
Spotlight Command Palette	`Ctrl + K` or `/`	Keyboard-first overlay for instant global search across all engines, jumping to any page, detonating sample strains, and controlling dashboard modes without using the mouse.
Interactive DEFCON Controller	Topbar Dropdown	Live switcher between DEFCON 1 (Emergency alert with pulsing crimson screen vignette) down to DEFCON 5 (Normal defense). Synchronizes ambient lighting and sound sirens.
Dynamic Theme Accents	Topbar Theme Menu	Switches color themes instantly: Cyber Cyan, Crimson Bloodline, Matrix Emerald, and Cyberpunk Violet. Updates CSS variables dynamically and persists in localStorage.
AI SOC Defense Copilot	`Alt + C` / Floating FAB	Interactive AI analyst assistant slide-out drawer providing incident triage playbooks, threat actor profiling (APT29, Volt Typhoon), CVE briefings, and instant firewall drop rules.
Enterprise MITRE ATT&CK; Matrix	Dashboard Grid	10-column interactive heatmap displaying real-time TTP density (Initial Access to Impact). Clicking any technique opens detection signatures and links to correlation graphs.
Tactical Web Audio FX	`Alt + M`	Procedural Web Audio API sound synthesizer producing tactile UI clicks, sonar radar chirps on missile intercepts, and two-tone sirens on DEFCON 1 alerts without external audio files.
War Room HUD CRT Mode	`Alt + W`	Overlays vintage CRT phosphor scanlines and tactical HUD corner brackets on cards for high-contrast war-room monitors and conference demonstrations.
Keyboard Shortcut Cheat-Sheet	`?`	Quick-reference modal displaying all two-key jump shortcuts (G+D for Dashboard, G+M for Map, G+F for Feed, G+I for IOC Search, G+C for Correlation).

4. DATABASE PERSISTENCE & RELATIONAL ENTITY SCHEMA

Entity Model	Primary Fields & Types	Purpose & Business Logic
User	id (PK), email, password_hash, name, role, department, badge_id, clearance_level, last_login	SOC operator identity, role-based access control (Analyst, Threat Hunter, CISO), PBKDF2 password hashing.
SecurityAlert	id (PK), title, severity, category, source_ip, target_ip, status, assigned_to, analyst_notes, created_at	Operational SOC alert triage queue with state tracking (NEW, INVESTIGATING, CONTAINED, CLOSED).
ThreatLog	id (PK), event_id, source_ip, destination_ip, protocol, threat_type, action_taken, signature, timestamp	High-velocity telemetry event log recording perimeter drops, sensor intercepts, and traffic volume.
ThreatIndicator	id (PK), indicator_value, indicator_type, threat_actor, confidence, tlp, first_seen, last_seen	Curated indicator repository backing the STIX/TAXII feed, correlation engine, and OSINT lookup.

InvestigationCase	id (PK), case_number, title, lead_investigator, status, mitre_ttp, priority, evidence_json	Forensic case dossiers tracking incident response lifecycle and multi-indicator correlation graphs.

5. API SPECIFICATIONS & TELEMETRY CONTRACTS

Endpoint	HTTP Method	Authentication	Payload & Response Description
`/api/live-attack`	GET	Session Required	Returns real-time attack telemetry event with IST timestamp, source/target geo coordinates, MITRE codes, and hex byte preview.
`/api/live-attacks-batch`	GET	Session Required	Returns batch of N (default 8, max 25) real-time events for instant map and terminal initial seeding.
`/global-search?q=...`	GET	Session Required	Smart query classification dispatcher routing IPs/Domains to IOC Search, CVEs to CVE Explorer, and Actors to Correlation Engine.
`/api/alerts/<id>/status`	POST	Session Required	Updates security incident status, logs analyst note with timestamp, and assigns investigator.
`/api/export/stix`	POST	Session Required	Generates and returns an OASIS STIX 2.1 JSON bundle with attack-pattern and indicator objects.
`/export/report/json`	GET	Session Required	Exports full executive CISO briefing dataset in structured JSON format.
`/export/report/csv`	GET	Session Required	Streams raw threat feed records as downloadable comma-separated values (CSV).

6. SERVERLESS CLOUD DEPLOYMENT & CI/CD PIPELINE

Deploying Flask applications to serverless runtimes like Vercel introduces unique challenges, specifically the **read-only root filesystem** (`/var/task`) and request routing. CTICD solves this with zero external dependencies through two dedicated configuration files:

1. Serverless Routing Configuration (`vercel.json`):

Directs Vercel to use the `@vercel/python` builder, serves static assets (`/static/*`) directly via Vercel's global Edge CDN, and routes dynamic requests through the WSGI gateway.

2. Serverless WSGI Entrypoint (`api/index.py`):

Ensures project root directories are in the system path and exports the Flask `app` instance as the callable serverless handler.

3. Self-Healing SQLite /tmp Adaptation (`config.py`):

Detects the `VERCEL=1` environment flag at runtime. If running on Vercel, it copies the seeded SQLite database (`instance/cticd.db`) into `/tmp/cticd.db`, the only writable filesystem location on AWS Lambda/Vercel serverless containers. This ensures user logins, session cookies, and alert status mutations function without read-only filesystem crashes.

7. SECURITY, GOVERNANCE & COMPLIANCE CONTROLS

• **Authentication Hardening:** Passwords hashed using PBKDF2 with SHA-256 and unique per-user salts. Session cookies protected with `HttpOnly` and `SameSite=Lax` flags.
• **Injection Defense:** All database queries execute via SQLAlchemy parameterized ORM objects, eliminating SQL injection vectors. All user inputs are strictly sanitized and cast.
• **Traffic Light Protocol (TLP 2.0):** Curated threat indicators are tagged with standardized TLP markings (TLP:CLEAR, TLP:GREEN, TLP:AMBER, TLP:AMBER+STRICT, TLP:RED).
• **STIX 2.1 & TAXII Standards:** Exporter serializes indicator bundles conformant to OASIS CTI technical committee specifications.

8. CONCLUSION & FUTURE ROADMAP

CTICD v2.5 delivers a high-impact, production-ready cyber threat intelligence and operations command platform. By fusing millisecond-precision geospatial trajectory modeling, Diamond Model intrusion graph analysis, PE entropy binary dissection, and automated defense synthesis into a unified interface, CTICD reduces analyst Mean Time to Detect (MTTD) and Mean Time to Respond (MTTR).

Future Strategic Enhancements:
1. Distributed real-time event streaming via Apache Kafka or Redis Pub/Sub for millions of events per second.
2. Bi-directional SIEM connectors for automatic alert ingestion from Splunk, Elastic Security, and Microsoft Sentinel.
3. Automated SOAR containment integrations triggerable directly from the AI Defense Copilot.